From Daily to Hourly Marine Heatwaves: A Taxonomy-Based, GPU-Accelerated Framework for Event Detection and Classification

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ABSTRACT

Marine heatwaves (MHWs) are traditionally identified from daily Sea Surface Temperature (SST) records, yet this framework does not transfer directly to higher temporal resolution. At hourly scale, SST reveals sub-daily variability that is largely smoothed out in daily means, including intermittent threshold exceedances, short-lived interruptions, diurnal cycling, and nocturnal heat retention. These features complicate the definition of event continuity and persistence, but also provide an opportunity to examine the internal structure of marine heat extremes in greater detail. This study presents a methodology for detecting and analysing MHWs at hourly resolution. Events are identified through a duration–gap framework and subsequently decomposed into individual diurnal cycles, enabling a taxonomy based on intra-event morphology. This allows MHWs to be characterised not only by duration and intensity, but also by persistence, cycle-to-cycle evolution, and sub-daily dynamics that are not accessible from daily data. A sensitivity analysis is used to assess the robustness of the framework by systematically varying key methodological parameters and evaluating their influence on event identification, continuity, and internal structure. The methodology is tested using in situ hourly SST observations from the Dragonera buoy in the Balearic Islands, which provide a high-frequency observational reference for evaluating the temporal behaviour of detected events. The proposed approach is intended as a zoom-in diagnostic layer rather than a replacement for standard daily MHW detection: daily methods can first be used to identify relevant or extreme events, after which selected time windows can be re-analysed at hourly resolution to resolve their internal evolution. As a next step, the resulting SST-based event catalogue will be combined with high-frequency coastal air-temperature observations from Cyprus to investigate compound marine–atmospheric heat extremes and their role in amplifying coastal thermal stress, thereby outlining the conceptual traits of a so-called "thermal cage".

1. Introduction

Some Introduction

2. Architecture

Some Architecture

3. Methodology

Some methodology

4. Results

Some Results

5. Conclusions

Some conclusions

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Some Appendix

References